19. There are four of them; call them X, Y, Z, and $\mathbb{Z}_6$ corresponding respectively to subgroups $\mathbb{Z}_6, \langle 2\rangle, \langle 3\rangle$, and $\{0\}$. See the text answer for the action tables for X, Y, and Z. The group table for $\mathbb{Z}_6$ is essentially the action table for $\{0\}$. (Conceptually, the entries in the body of that group table should in braces, like $\{3\}$, to denote the coset rather than the element.)

20. There are four of them; using the notation for S_3 in Section 8, call them X, Y, Z, and S_3 corresponding respectively to the subgroup S_3, the subgroup $\{\rho_0, \rho_1, \rho_2\}$, the three conjugate subgroups $\{\rho_0, \mu_1\}, \{\rho_0, \mu_2\}$, and $\{\rho_0, \mu_3\}$, and the trivial subgroup $\{\rho_0\}$. We choose the subgroup $\{\rho_0, \mu_1\}$ to illustrate the action on the 3-element set. The action tables for these subgroups are

	X	Y		Z		
	a	a	b	a	b	c
ρ_0	a	a	b	a	b	c
ρ_1	a	a	b	b	c	a
ρ_2	a	a	b	c	a	b
μ_1	a	b	a	a	c	b
μ_2	a	b	a	c	b	a
μ_3	a	b	a	b	a	c

and essentially the group table for S_3 itself corresponding to the subgroup $\{\rho_0\}$. (Conceptually, the entries in the body of that group table should in braces, like $\{\mu_1\}$, to denote the coset rather than the element.)

17. Applications of G-Sets to Counting

1. $G = \langle (1,3,5,6)\rangle$ has order 4. Let $X = \{1,2,3,4,5,6,7,8\}$. We have $|X_{(1)}| = 8$, $|X_{(1,3,5,6)}| = |\{2,\ 4,\ 7,\ 8\}| = 4$, $|X_{(1,5)(3,6)}| = |\{2,\ 4,\ 7,\ 8\}| = 4$, $|X_{(1,6,5,3)}| = |\{2,\ 4,\ 7,\ 8\}| = 4$. Therefore we have $\sum_{g\in G} |X_g| = 8+4+4+4 = 20$. The number of orbits under G is then $(1/4)(20) = 5$.

2. The group G generated by $(1,\ 3)$ and $(2,\ 4,\ 7)$ has order 6. Let $X = \{1,2,3,4,5,6,7,8\}$. We have

$$\begin{array}{lll} |X_{(1)}| = 8 & |X_{(1,3)}| = 6 & |X_{(2,4,7)}| = 5 \\ |X_{(2,7,4)}| = 5 & |X_{(1,3)(2,4,7)}| = 3 & |X_{(1,3)(2,7,4)}| = 3. \end{array}$$

Thus $\sum_{g\in G} |X_g| = 8+6+5+5+3+3 = 30$. The number of orbits under G is then $(1/6)(30) = 5$.

3. The group of rigid motions of the tetrahedron has 12 elements because any one of four triangles can be on the bottom and the tetrahedron can then be rotated though 3 positions, keeping the same face on the bottom. We see that $|X_g| = 0$ unless g is the identity ι of this group G, and $|X_\iota| = 4! = 24$. Thus there are $(1/12)(24) = 2$ distinguishable tetrahedral dice.

4. The total number of ways such a block can be painted with different colors on each face is $8\cdot 7\cdot 6\cdot 5\cdot 4\cdot 3$. The group of rigid motions of the cube has 24 elements. The only rigid motion leaving unchanged a block with different colors on all faces is the identity, which leaves all such blocks fixed. Thus the number of distinguishable blocks is $(1/24)(8\cdot 7\cdot 6\cdot 5\cdot 4\cdot 3) = 8\cdot 7\cdot 5\cdot 3 = 40\cdot 21 = 840$.

5. There are 8^6 ways of painting the faces of a block, allowing for repetition of the 8 colors. Following the breakdown of the group G of rotations given in the hint, and using sublabels to suggest the categories in this breakdown, we have
$|X_\iota| = 8^6$ where ι is the identity,
$|X_{\text{opp face, 90° or 270° rotation}}| = 8 \cdot 8 \cdot 8$; there are 6 such,
$|X_{\text{opp face, 180° rotation}}| = 8 \cdot 8 \cdot 8 \cdot 8$; there are 3 such,
$|X_{\text{opp vertices}}| = 8 \cdot 8$; there are 8 such,
$|X_{\text{opp edges}}| = 8 \cdot 8 \cdot 8$; there are 6 such.
Thus $\sum_{g \in G} |X_g| = 8^6 + 6 \cdot 8^3 + 3 \cdot 8^4 + 8 \cdot 8^2 + 6 \cdot 8^3 = 8^3(8^3 + 37)$. The number of distinguishable blocks is thus $(1/24)[8^3(8^3 + 37)] = 11,712$.

6. Proceding as in Exercise 5 using the same group G acting on the set X of 4^8 ways of coloring the eight vertices, we obtain
$|X_\iota| = 4^8$ where ι is the identity,
$|X_{\text{opp face, 90° or 270° rotation}}| = 4 \cdot 4$; there are 6 such,
$|X_{\text{opp face, 180° rotation}}| = 4 \cdot 4 \cdot 4 \cdot 4$; there are 3 such,
$|X_{\text{opp vertices}}| = 4 \cdot 4 \cdot 4 \cdot 4$; there are 8 such,
$|X_{\text{opp edges}}| = 4 \cdot 4 \cdot 4 \cdot 4$; there are 6 such.
Thus $\sum_{g \in G} |X_g| = 4^8 + 6 \cdot 4^2 + 3 \cdot 4^4 + 8 \cdot 4^4 + 6 \cdot 4^4 = 4^4(273) + 96$. The number of distinguishable blocks is thus $(1/24)[4^4(273) + 96)] = 2,916$.

7. **a.** The group is $G = D_4$ and has eight elements. We label them as in Section 8. There are $6 \cdot 5 \cdot 4 \cdot 3$ ways of painting the edges of the square, and we let X be this set of 360 elements. We have $|X_{\rho_0}| = 6 \cdot 5 \cdot 4 \cdot 3$ and $|X_g| = 0$ for $g \in D_4, g \neq \rho_0$. Thus the number of distinguishable such painted squares is $(1/8)(360) = 45$.

b. We let G be as in Part(**a**), and let X be the set of 6^4 ways of painting the edges of the square, allowing repetition of colors. This time, we have

$$|X_{\rho_0}| = 6^4, \;\; |X_{\rho_1}| = |X_{\rho_3}| = 6, \;\; |X_{\rho_2}| = 6 \cdot 6,$$
$$|X_{\mu_1}| = |X_{\mu_2}| = 6 \cdot 6 \cdot 6, \;\text{ and }\; |X_{\delta_1}| = |X_{\delta_2}| = 6 \cdot 6.$$

Thus $\sum_{g \in G} |X_g| = 6^4 + 2 \cdot 6 + 6^2 + 2 \cdot 6^3 + 2 \cdot 6^2 = 6^2(51) + 12$. The number of distinguishable blocks is thus $(1/8)[6^2(51) + 12)] = 231$.

8. The group of rigid motions of the tetrahedron is a subgroup G of the group of permutations of its vertices. The order of G is 12 because, viewing the tetrahedron as sitting on a table, any of the four faces may be on the botton, and then the base can be rotated repreatedly through 120° to give three possible positions. If we call the vertex at the top of the tetrahedron number 1 and number the vertices on the table as 2, 3, and 4 counterclockwise when viewed from above, we can write the 12 group elements in cyclic notation as

#1 on top	#2 on top	#3 on top	#4 on top
(1)	(1, 2)(3, 4)	(1, 3)(2, 4)	(1, 4)(2, 3)
(2, 3, 4)	(1, 3, 2)	(1, 2, 3)	(1, 2, 4)
(2, 4, 3)	(1, 4, 2)	(1, 4, 3)	(1, 3, 4).

Let X be the 2^6 ways of placing either a 50-ohm resistor or 100-ohm resistor in each edge of the tetrahedron. Now the elements of G that are 3-cycles correspond to rotating, holding a single vertex fixed. These carry the three edges of the triangle opposite that vertex cyclically into themselves, and